

Practice Problems for Week 3

Operating and Systems Programming 2022/23

For all questions involving code, try to answer them *without* running the code first.

1. What is a function pointer in C? Give an example where function pointers may be useful.
2. What is the difference between `int *foo (int arg)` and `int (*foo) (int arg)?`
3. Consider the following program:

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]){
    int i;

    printf ("Have %d arguments\n", argc);
    for (i = 0; i < argc; i++) {
printf ("argument %d is %s\n", i, argv[i]);
    }
    return 0;
}
```

- What is the output of this program when called with
`./program arg1 arg2 arg3`
 - What is the output of this program when called with
`./program First argument Second argument`
 - What is the output of this program when called with
`./program "First argument" "Second argument"`
4. What is the purpose of a void pointer? Where do you need it?
 5. Why is it important that void pointers are cast to the correct type? Give an example which demonstrates this.

6. What is the purpose of a Makefile?
7. Consider the following Makefile

```
CFLAGS=-Wall -Werror -g

TARGETS = program

all: program

program: program.o input.o output.o
    $(CC) $(CFLAGS) -o $@ input.o output.o $<

%.o: %.c
    $(CC) -c $(CFLAGS) $<

clean:
    rm -f $(TARGETS) *.o
```

and assume the directory listing is

```
-rw-r--r-- 1 exr joint 202 Oct 6 16:28 input.c
-rw-r--r-- 1 exr joint 203 Oct 6 16:29 Makefile
-rw-r--r-- 1 exr joint 74 Oct 6 16:26 output.c
-rw-r--r-- 1 exr joint 6032 Oct 11 13:32 output.o
-rw-r--r-- 1 exr joint 183 Oct 6 16:26 program.c
```

Which commands are executed when the command `make` is given?